Rapid Revision Summary

Last-minute review · condensed from 16 papers and mark schemes, Jan 2019 – Jun 2026

THE PAPER

1 h 30 min · 100 marks · 25–29 short questions · calculator · under 1 min per mark. **No formulae sheet.** Includes matrices, loci, symmetry and formal proofs.

LEARN (NOT GIVEN)

Cone $\frac{1}{3}\pi r^2 h$, curved πrl · sphere $\frac{4}{3}\pi r^3$, $4\pi r^2$ · cylinder $\pi r^2 h$, $2\pi rh$ · sector $\theta/360 \times \pi r^2$, arc $\theta/360 \times 2\pi r$

Sine rule · cosine rule · $\frac{1}{2}ab \sin C$ · quadratic formula · Pythagoras · SOH CAH TOA

TOP 12 MARK-LOSERS

- Surd/fraction answer without the working = **0**.
- Numerical examples used as a "proof".
- Congruence pairs without reasons or condition.
- Angle reasons missing the key words.
- Construction without visible arcs.
- Region R shaded with a wrong locus.
- Matrix unknowns not written as equations.
- Rearranged formula not starting "x = …".
- Reverse % done with × 0.96 instead of ÷ 1.04.
- Wrong bound in a quotient.
- Chord used instead of a tangent for a gradient.
- Probability > 1 in working.

NUMBER

SURDS

$(\sqrt{12} + \sqrt{300})/\sqrt{27} = 12\sqrt{3}/3\sqrt{3} = 4$ · $3\sqrt{180} - 2\sqrt{245} = 4\sqrt{5} = \sqrt{80} \cdot \sqrt{700} - \sqrt{343} = 3\sqrt{7} = \sqrt{63}$

$(\sqrt{13} + \sqrt{11})/(\sqrt{13} - \sqrt{11}) = (24 + 2\sqrt{143})/2 = 12 + \sqrt{143}$

- LCM(18, 30, 48) = 720 · HCF(216, 540) = 108.
- Remainder 3 on ÷28 and ÷120 ⇒ LCM + 3 = 843.
- $2^{36} - 1 = (2^{18} - 1)(2^{18} + 1)$.
- $1/p = p - 1$ for $p = (1 + \sqrt{5})/2$ by rationalising.
- Mixed numbers: improper → common denominator → simplest mixed number.

PERCENTAGES, RATIO, PROPORTION

- +4% → \$3.64 ⇒ ÷ 1.04 = 3.50 ⇒ rise \$0.14.
- \$27 = 108% ⇒ cost 25 ⇒ 35% profit price 33.75.
- a : b = 3 : 5, a : c = 7 : 4 ⇒ 21 : 35 : 12.
- $f = kg^2$, $g = m/h$ ⇒ $f = 144/h^2$.
- $y \propto 1/\sqrt[3]{x}$: $y = k/\sqrt[3]{x}$.

BOUNDS

$\text{Speed}_{max} = d_{max}/t_{min}$: 215/3.375 = 63.7 km/h.
$\text{Time}_{max} = d_{max}/v_{min}$: 192.5/61.5 = 3 h 8 min.
$(P - Q)_{max} = P_{max} - Q_{min}$.

ALGEBRA

- $2xy(4y - 9) \cdot 7x^2y^2(2y - 3x^3)$: factorise **completely**.
- $(125p^9w^3)^{2/3} = 25p^6w^2$.
- $(3x^2 - 8x - 3)/(4x^2 - 36) = (3x + 1)/4(x + 3)$.
- $15 - 28x - 7x^2 = 43 - 7(x + 2)^2$.
- $9 - 10x - 5x^2 = 14 - 5(x + 1)^2$ ⇒ range g(x) ≤ 14.
- Factor theorem: f(5) = 0 ⇒ (x − 5)(2x − 3)(x + 2).
- (2n + 1)(n + 1)/2(n + 1) = (2n + 1)/2: odd ÷ 2 ⇒ never an integer.

EQUATIONS AND INEQUALITIES

- Fractions: multiply **every** term by the LCM.
- $6x^2 + 5x - 25 > 0$ ⇒ $(3x - 5)(2x + 5) > 0$ ⇒ $x < -2.5$ or $x > 5/3$.
- $-2x + 4 \le 3x - 6 < 2x + 2$ ⇒ $2 \le x < 8$ (closed ●, open ○).
- Regions: equation of each line → three inequalities.
- $f(x) = x^2 - 2$ ⇒ range $f(x) \ge -2$.

MATRICES

TOOLKIT

AB: row × column; order m×n × n×p.

det(a b / c d) = ad − bc · inverse = (1/det)(d −b / −c a).

(4 3 / −2 1)(−7 −2 / 1 3) = (−25 1 / 15 7).

Unknowns: 12 + 4p = 4 ⇒ p = −2 (equation outside the matrix).

GRAPHS AND CALCULUS

- $m = \Delta y/\Delta x$; ⊥ gradient $-1/m$.
- $y = 2x^4 - 2/x$ ⇒ $dy/dx = 8x^3 + 2/x^2$.
- Speed–time: gradient = acceleration; area = distance.
- Curve gradient: draw a **tangent**.
- v = ds/dt, a = dv/dt; rest when v = 0.

PROOFS

CONGRUENCE (3)

Three equal pairs, each with a reason, + SSS / SAS / ASA / AAS / RHS.

BQ = QC (midpoint); PQ = RC, QR = BP (opposite sides of a parallelogram) ⇒ SSS.

SIMILARITY (3)

Two angle pairs with reasons: same segment + vertically opposite ⇒ similar.

CIRCLE THEOREMS

Centre = 2 × circumference · semicircle 90° · same segment equal · cyclic 180° · tangent ⊥ radius · equal tangents · alternate segment.

Secants: EA × EB = ED × EC ⇒ 7.5(7.5 + x) = 84 ⇒ x = 3.7.

- Exterior angle = 360/n: 9° ⇒ 40 · interior 176° ⇒ 90 · sum 8280° ⇒ n = 48.

CONSTRUCTIONS AND LOCI

Locus	Construct
distance from point	arc/circle (scale!)
equidistant from 2 points	perpendicular bisector, 2 pairs of arcs
equidistant from 2 lines	angle bisector
distance from a line	parallel line

- "Closer to B than C": B's side of the bisector.
- R: all loci correct, shade, label R.
- Symmetry: lines vs rotational order — check both.

MENSURATION AND SIMILARITY

- Length k · area k^2 · volume k^3.
- Volume +33.1% ⇒ k = 1.1 ⇒ area × 1.21. Volume −43% ⇒ $k^3 = 0.57$.
- EB/DC = AE/AD ⇒ EB = 9, AC = 11.9.
- Composite solids: add/subtract parts.

TRIG AND VECTORS

- Elevation from a raised point: use the height difference.
- Smallest angle: opposite the shortest side (cosine rule).
- 3D: half the base diagonal for angle with the base.
- Bearings: draw north lines; cosine then sine rule.
- $|(x, y)| = \sqrt{(x^2 + y^2)}$; a = kb ⇒ parallel; equate coefficients for ratios.

STATISTICS AND PROBABILITY

- Mean = Σxf ÷ Σf; missing x from a given mean: Σxf = mean × Σf.
- Combined: total = mean × n.
- Frequency density = f ÷ width.
- Probabilities sum to 1 → find x first.
- Only one = P(W, not) + P(not, W). Not same = 1 − P(same).
- n(R ∩ C′) = R only; Venn from the centre.